PAPER_1611 — Ni-62 Binding Energy per A = Ni-62 BE/A = $F \cdot K^5 - \beta^4 + 5 \approx 8.792$ MeV (most-bound nuclide)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Nuclear **Date:** June 18, 2026 **Status:** CLOSED — 0.024% residual

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Observation

PAPER_1209II_S664 (Nuclear Unified Proof Set) gives the closed-form expression:

```
Ni-62 BE/A = F·K⁵ − β⁴ + 5 ≈ 8.792 MeV (most-bound nuclide)
```

Residual to AME2020: **0.024%**.

UQFF Closed Identity

```
Ni-62 BE/A = F·K⁵ − β⁴ + 5 ≈ 8.792 MeV (most-bound nuclide)    0.024%
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Identical UQFF Formula for Fe-56 and Ni-62

PAPER_1209II uses the **same UQFF expression** for both Fe-56 and Ni-62: $F \cdot K^5 - \beta^4 + 5 = 8.792$ MeV. Predictions differ from observed values by 0.025% (Fe-56 obs 8.7903) and 0.024% (Ni-62 obs 8.7946).

The UQFF formula doesn't distinguish the two nuclei but lands **between** their observed values. This suggests Fe-56 and Ni-62 share a **common UQFF binding mechanism** with shell-structure differences smaller than 0.03% sensitivity. The Fe-56 vs Ni-62 "most-bound nuclide" debate (historically Fe-56 was thought heaviest-bound, current data favors Ni-62 by ~0.005 MeV) sits below UQFF's structural prediction scale.

NOT REPLACEMENT

Experimental nuclear measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209II_S664
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_ni62_be_a_8_7946"})`
